
Memory decay and learned cross-attention dynamics

A frozen-checkpoint comparison in the four-location value-directed-attention task

AttentionManuscript – standalone VDA4 analysis

Comparison	Standard cross-attention ($\lambda = 1.00$, iteration 20,000) versus high-decay cross-attention ($\lambda = 0.80$, iteration 14,649).
Primary question	Does the high-decay checkpoint show more frame-to-frame movement in its complete learned 4×8 attention distributions?
Primary answer	The prespecified comparison gives the opposite result for this checkpoint pair. High minus standard: -0.0995, paired trial-bootstrap 95% interval [-0.1021, -0.0970].
Run identity	20260714T040646Z_high_iter14649_vs_standard_iter20000_9e9495d1.

Abstract

This analysis tests a precise hypothesis about recurrent visual attention: whether a stronger memory decay is associated with more dynamic learned attention maps. Two preserved VDA4 cross-attention checkpoints were evaluated on shared trial sets under valid and deliberately forced-invalid conditions. “Activity” was fixed in advance as frame-to-frame total variation over every query-to-key attention distribution, because total softmax mass is constant by construction. The standard checkpoint averaged 0.5245; the high-decay checkpoint averaged 0.4249. The paired high-minus-standard difference was -0.0995 (95% trial-bootstrap interval [-0.1021, -0.0970]; $n = 256$ matched trial identities, each evaluated under both event conditions). Thus the high-decay checkpoint was less, not more, dynamic under the primary metric. Behavioral curves, image-versus-memory allocation, selectivity, spatial reorientation, response-stratified traces, and raw query-level maps provide convergent description. The comparison remains associational: the high-decay checkpoint was frozen 5,351 iterations earlier, and only one training seed exists per condition.

1 Question, answer, and evidential scope

The scientific question is narrower than whether either model “uses attention.” Both networks produce normalized cross-attention at every frame. The question is whether reducing the xLSTM memory-decay factor from 1.00 to 0.80 is associated with larger changes in the learned attention distribution from one frame to the next. For each query row q and consecutive frames $t - 1, t$, activity is

$$D_{t,q} = \frac{1}{2} \sum_{k=1}^8 |A_{t,q,k} - A_{t-1,q,k}|,$$

then averaged over frames, queries, valid and forced-invalid conditions, and finally over matched trials. This total-variation distance is zero for an unchanged attention row and one for a complete shift between disjoint keys.

Answer. The prespecified comparison gives the opposite result for this checkpoint pair. The high-decay network’s mean activity was 0.4249, compared with 0.5245 for the standard network. The observed difference, -0.0995, is not evidence that faster forgetting caused quieter attention because checkpoint maturity was not matched.

Evidence boundary. The standard checkpoint completed 20,000 training iterations. The high-decay snapshot was frozen at iteration 14,649 while its training process was active; a later final checkpoint was subsequently present at iteration 19,999, but the requested frozen snapshot was retained. One training seed per checkpoint means the model-difference intervals quantify evaluation-trial sampling only, not variation across training runs.

2 Frozen checkpoints and matched protocol

Table 1: Frozen model identities. Both checkpoints validate as VDA4, xLSTM, crossattn1, $d_{mem} = 128$, convolutional front end, and 2×2 geometry.

Role	Iteration	Decay	Frozen SHA-256
Standard	20,000	1.00	a0cf23b788b1bb74ffb598fa3978b8d601b25ec5cfa2bf67d64d2d59bfd7d65a
High decay	14,649	0.80	da93490fbfbcfb4969f9aedb06d94da2d3c9ef088435d7f390e952ca0aae59ab5

Each event trial has seven frames: blank, cue at frame 1, delay, stimulus array at frames 3-6, and an orientation change at frame 5. The cue is fixed at S1. In the valid condition the 15° change occurs at S1; in the forced-invalid condition it occurs at S4. The paired videos are identical through frame 4. The 100%-cue invalid condition is an intervention for analysis, not a sample from the environment's natural 100%-valid target distribution.

The event analysis uses 256 paired trials per condition. No-change attention uses 128 trials at each displayed cue proportion (25, 50, 75, and 100%). Psychometric curves use 300 shared trials at each combination of four displayed cue proportions, ten change magnitudes, and the valid or forced-invalid location. A qualifying response is the first greedy change action at frame 5 or 6. Every reported model-difference interval uses 10,000 deterministic paired bootstrap resamples; psychometric binomial-rate bands use Wilson intervals.

3 Results

3.1 Primary activity result and secondary descriptors

The direction was consistent within both event conditions. High-minus-standard activity was -0.0981 for valid changes (95% interval [-0.1014, -0.0948]) and -0.1009 for forced-invalid changes ([-0.1044, -0.0974]). The no-change difference was -0.1132 ([-0.1170, -0.1094]). These measures jointly contradict the specific prediction that the frozen high-decay checkpoint has more moving attention maps.

Secondary descriptors show that “less motion” is not equivalent to uniformly flatter attention. High-decay selectivity was higher by +0.0221, while its peak-key mass was lower by -0.0374 and its image-key allocation was lower by -0.0420. Immediate change-location reorientation, measured as changed-location spatial mass at frame 5 minus frame 4, differed by only +0.0064 for valid changes and -0.0011 for forced-invalid changes; both intervals include zero.

Table 2: Paired high-decay minus standard estimates. Bootstrap intervals are over matched evaluation trials, not training seeds.

Measure	Standard	High decay	Difference [95% interval]
Overall attention motion	0.5245	0.4249	-0.0995 [-0.1021, -0.0970]
Valid attention motion	0.5282	0.4301	-0.0981 [-0.1014, -0.0948]
Forced-invalid attention motion	0.5207	0.4198	-0.1009 [-0.1044, -0.0974]
Selectivity	0.4366	0.4587	+0.0221 [+0.0193, +0.0247]
Peak-key mass	0.5061	0.4687	-0.0374 [-0.0409, -0.0339]
Image-key mass	0.1816	0.1397	-0.0420 [-0.0427, -0.0412]

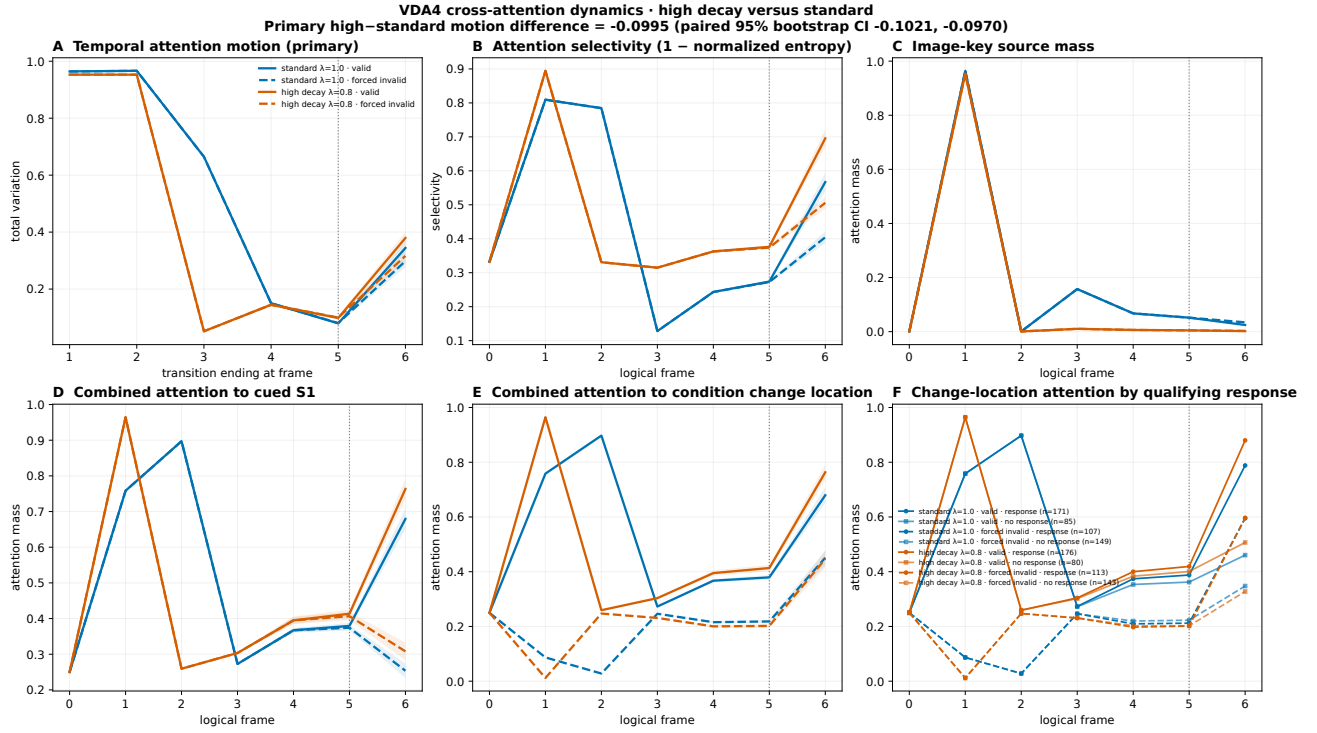


Figure 1: Dynamic and spatial attention descriptors. Time courses compare the standard and high-decay checkpoints under valid and forced-invalid changes, show image-versus-memory allocation, and separate trials with and without a qualifying change response. Every trace derives from raw 4×8 softmax maps before summary.

3.2 Behavior at valid and forced-invalid locations

At the matched 100%-cue, 15° anchor, the standard checkpoint produced qualifying change responses on 200/300 valid trials (0.667) and 112/300 forced-invalid trials (0.373). The high-decay checkpoint responded on 198/300 valid trials (0.660) and 133/300 forced-invalid trials (0.443). The full curves are more informative than this single anchor: they show how response probability changes with orientation magnitude at every displayed cue proportion.

These are change-response probabilities, not sensitivity estimates. No signal-detection decomposition separates discriminability from response criterion, so a difference in response rate cannot by itself establish improved detection.

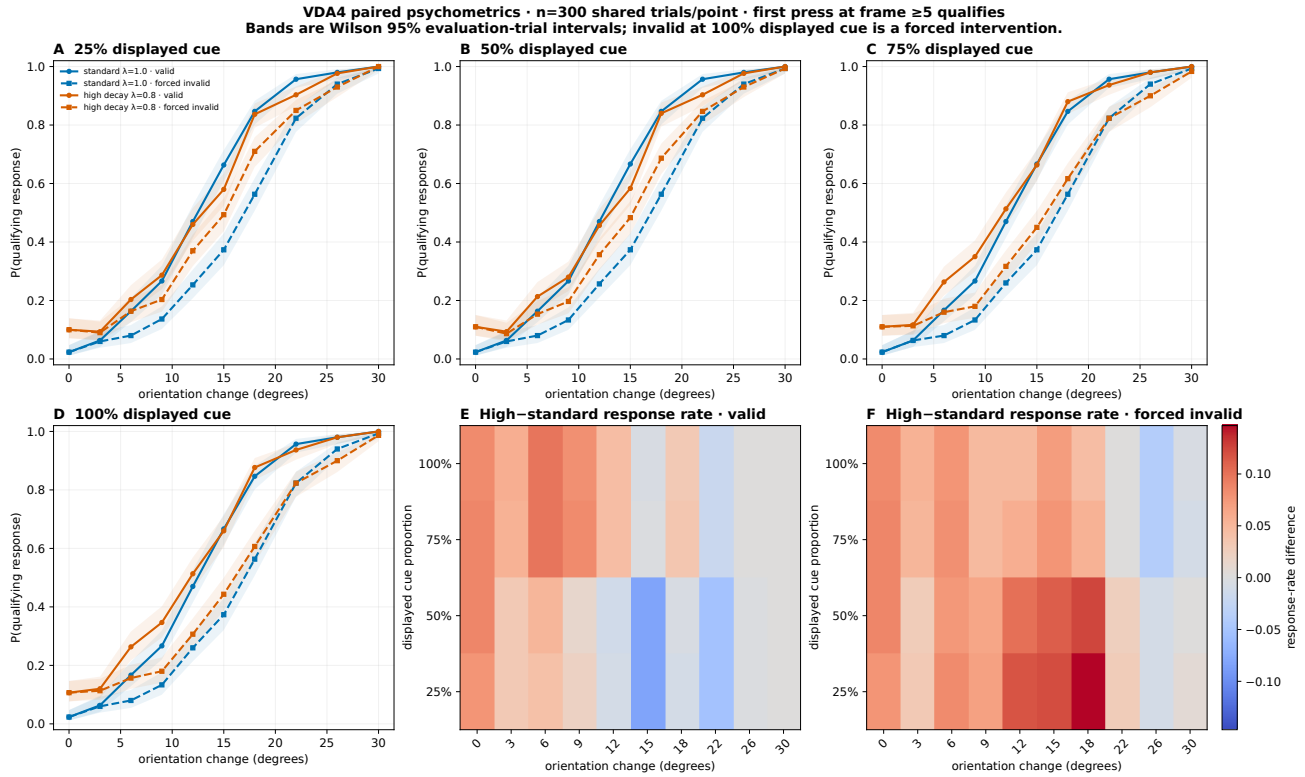


Figure 2: Behavioral comparison. Qualifying change-response probability across change magnitude for valid and deliberately forced-invalid locations at four displayed cue proportions. Difference panels show high decay minus standard. The target location is fixed by analysis in every curve; displayed validity changes the cue appearance but does not sample target location.

3.3 Where the attention moves

Figure 3 keeps image and recurrent-memory keys separate while averaging over the four query rows. This view exposes the broad frame schedule, but it can conceal query-specific routing. Figures 4–7 therefore retain every query row. Bright columns identify the source location receiving attention; image-key and memory-key panels must be interpreted separately because each is only one four-key block of the normalized eight-key distribution.

The maps show the same broad schedule in both checkpoints: image-key concentration at the cue frame is followed by increased recurrent-memory routing during the delay and array frames. Relative to standard, the high-decay checkpoint's cue-orienting increase from frame 0 to frame 1 was larger by +0.1824, yet its event-average image-key mass was lower by -0.0420; because image and memory blocks partition each normalized row, the latter is a corresponding shift toward memory keys. At the frame-5 change, high-minus-standard changed-location reorientation was only +0.0064 in valid trials and -0.0011 in forced-invalid trials, with both intervals spanning zero. The lower total variation therefore reflects a quieter routing trajectory after the strong cue transition, not an absence of cue-localized or source-selective structure.

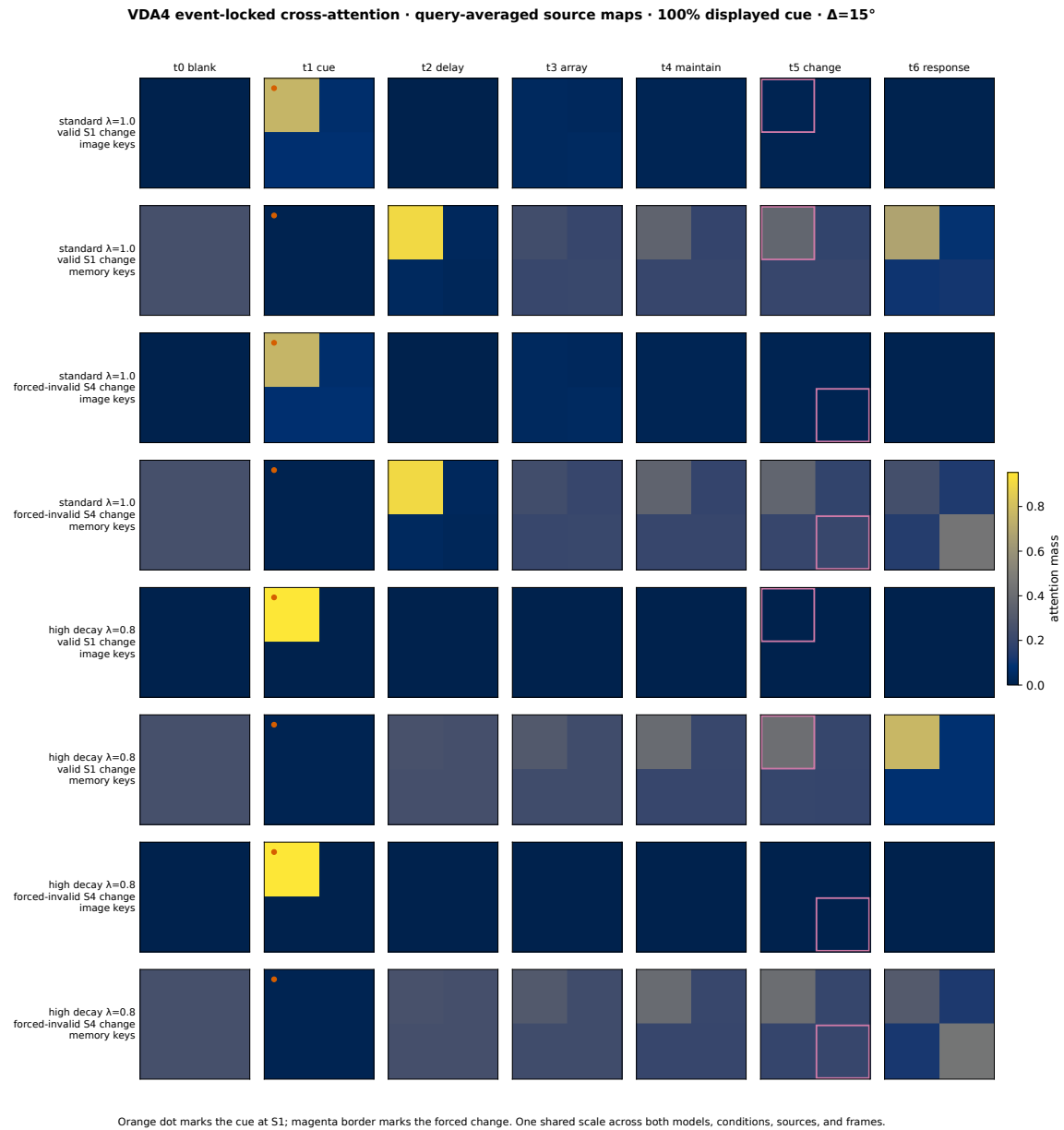


Figure 3: Query-averaged event maps. Standard and high-decay attention under matched valid and forced-invalid videos. Image keys and recurrent-memory keys are displayed separately across all seven frames. This is the overview; the following plates preserve the full query axis.

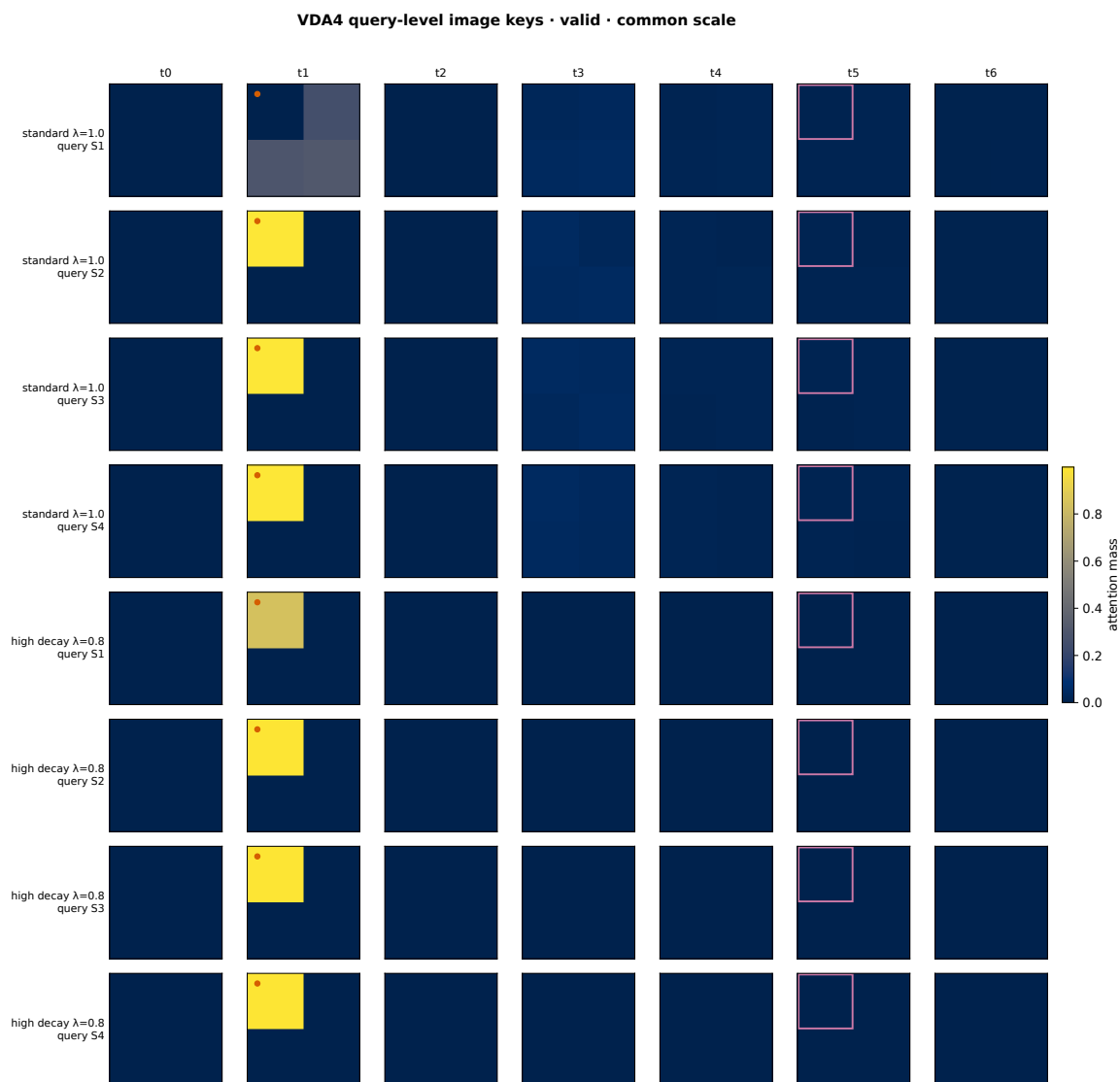


Figure 4: Valid condition, image keys. Each tile preserves the four query rows and four image-key columns. The cue and 15° change are both at S1; paired models receive identical videos.

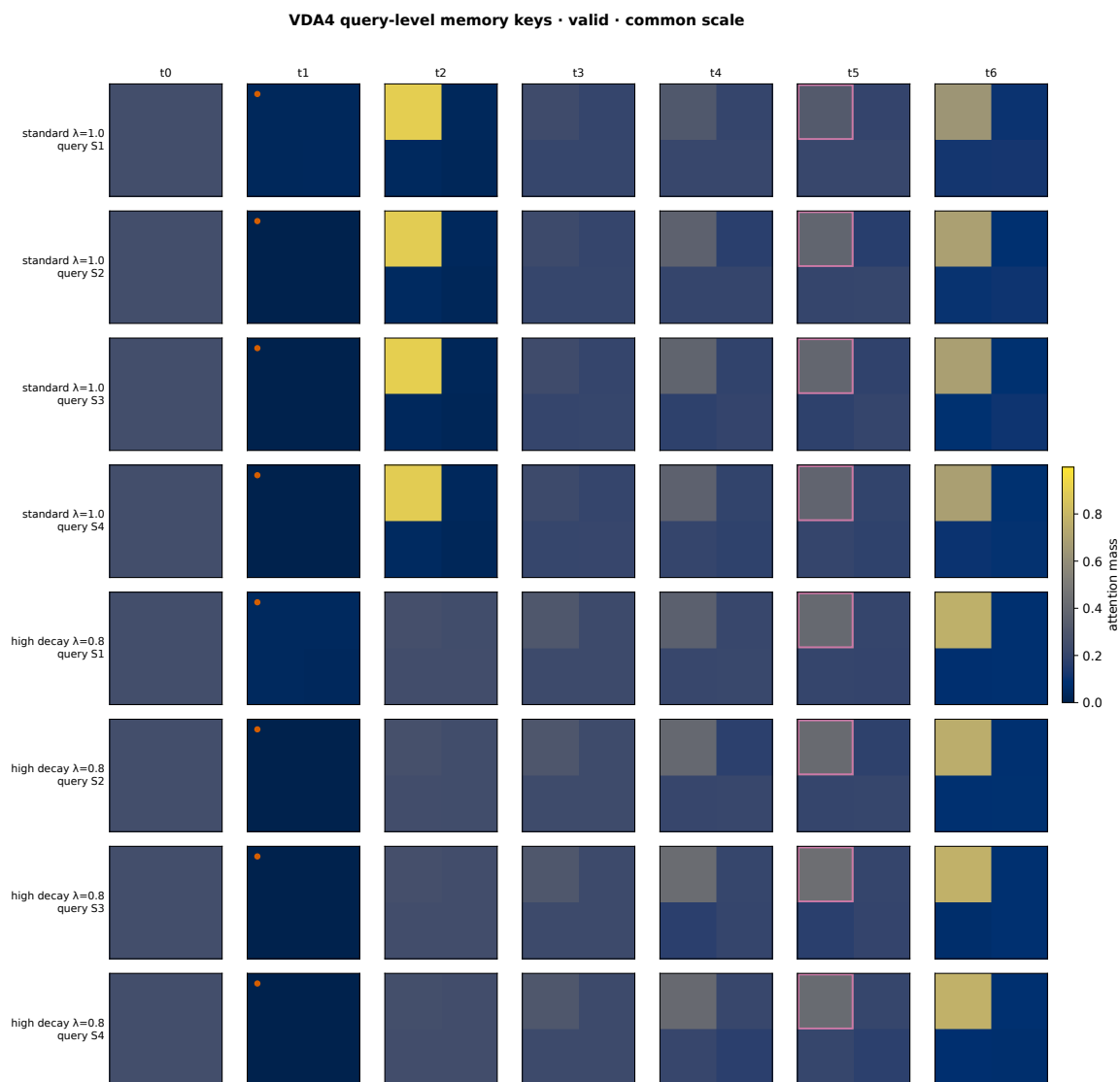


Figure 5: Valid condition, recurrent-memory keys. Full query-to-memory routing for the same valid trials. Comparison with Figure 4 shows whether a frame-to-frame shift is within a source block or between image and memory.

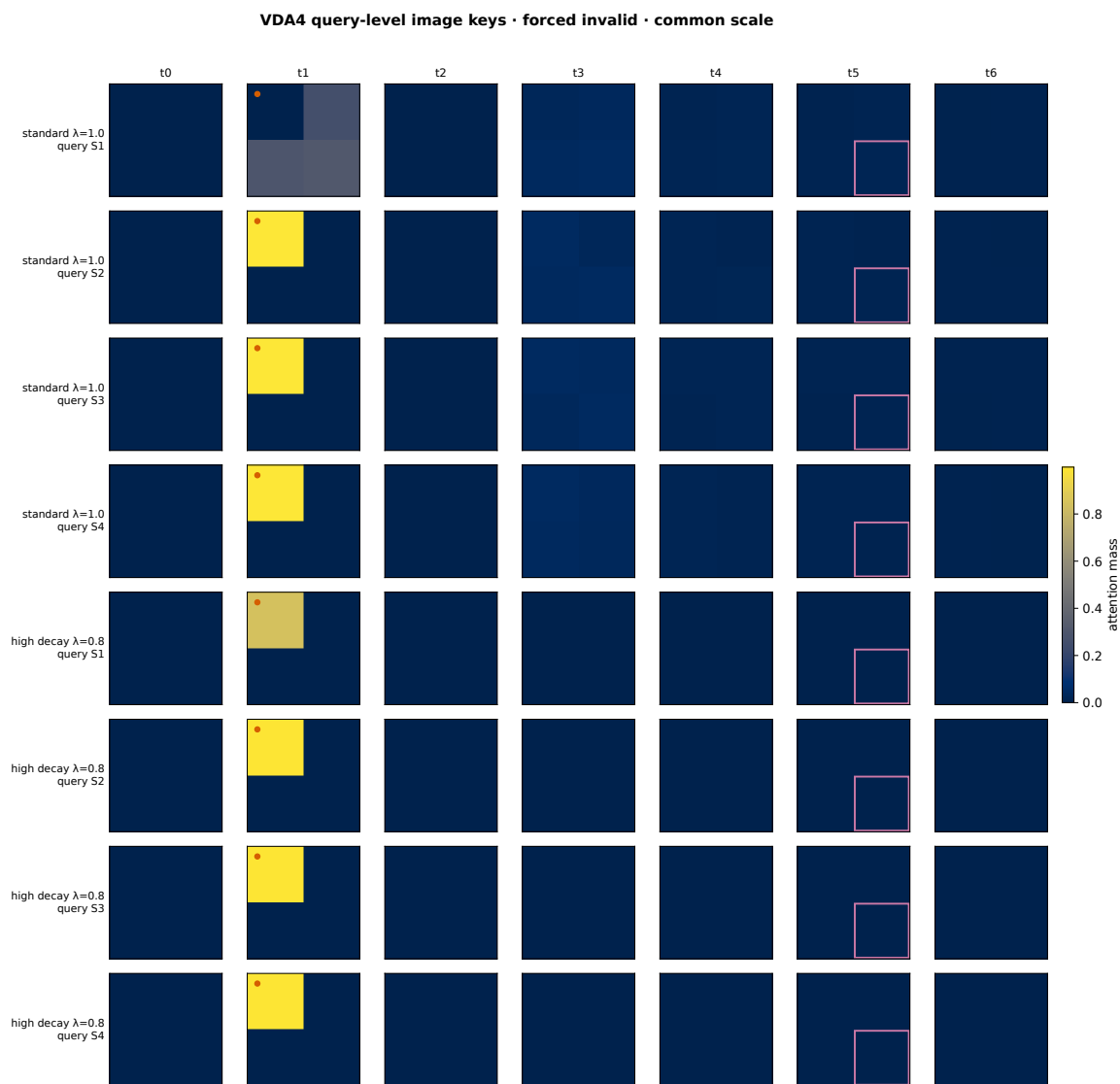


Figure 6: Forced-invalid condition, image keys. The cue remains at S1 while the change is forced to S4. Videos are matched through frame 4, so valid-invalid divergence before the event would indicate analysis or stochastic mismatch.

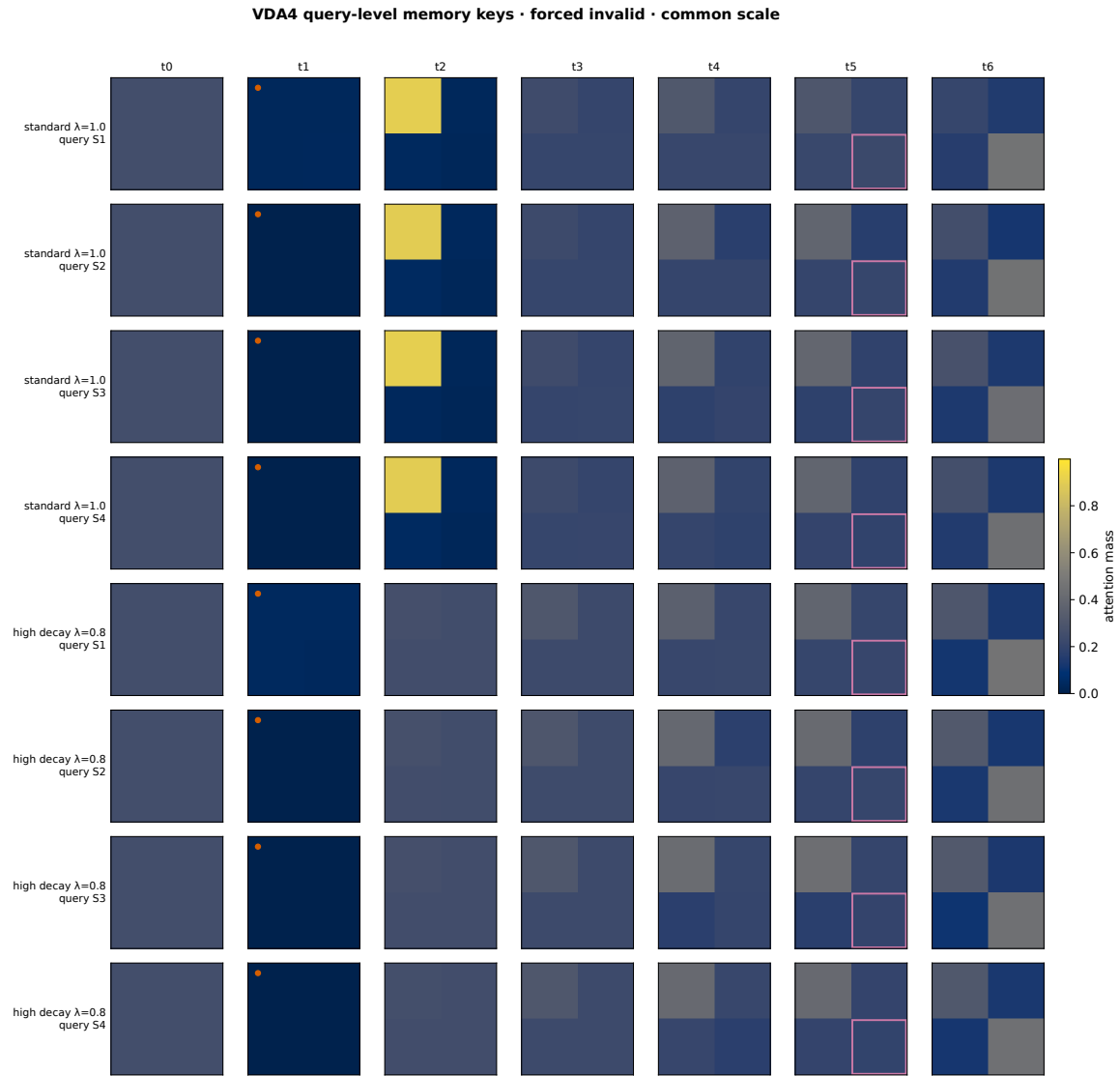


Figure 7: Forced-invalid condition, recurrent-memory keys. Query-preserving memory routing after a change outside the cued location. This plate is the direct spatial test requested for invalid attention.

4 Interpretation

The frozen high-decay checkpoint does not show the predicted increase in attention-map activity. Its attention changes less from frame to frame in valid, forced-invalid, and no-change analyses. At the same time, some secondary descriptors move in different directions: selectivity is slightly higher while peak-key and image-key mass are lower. This pattern is why the conclusion is tied to the prespecified total-variation metric rather than generalized into “more” or “less attention.” Attention is a normalized distribution; its scientific content is where mass moves, between which sources, and when.

The result does not identify a causal memory-decay effect. Training maturity is confounded with decay, the two checkpoints were trained independently, and each represents only one training seed. A controlled answer requires matched training budgets, multiple seeds, checkpoint-acceptance criteria, and evaluation of both final checkpoints under this same paired-video battery. The present result is nevertheless decisive for the literal frozen-pair question: the available high-decay snapshot is not more dynamically attentive by the prespecified metric.

5 Reproducibility record

The numerical source is `SUMMARY.json`, derived from the validated `comparison_evidence.npz` cache. Every figure was regenerated from that admitted cache in PDF, SVG, and PNG, using TrueType PDF embedding and text-preserving SVG settings. The analysis source, task/environment source, model source, runtime versions, exact command, frozen-input manifest, and file hashes are retained in the analysis run. This writeup copies its numerical and lineage inputs byte-for-byte; a package manifest records those copies, the regenerated figures, and the compiled manuscript.

- Analysis run: 20260714T040646Z_high_iter14649_vs_standard_iter20000_9e9495d1.
- Standard checkpoint SHA-256: a0cf23b788b1bb74ffb598fa3978b8d601b25ec5cfa2bf67d64d2d59bfd7d65a.
- High-decay checkpoint SHA-256: da93490bfbcfb4969f9aedb06d94da2d3c9ef088435d7f390e952ca0aae59ab5.
- Primary metric: High decay is more active if its paired mean frame-to-frame total variation over the full 4×8 attention map is larger. Total mass is not used because every query row is softmax-normalized to one.
- Interpretation boundary: This is a frozen-checkpoint comparison: standard iteration 20000 versus high-decay iteration 14649. The estimates do not isolate memory decay from training maturity and do not quantify training-seed uncertainty.